4. Functional Structure

4.1. Objective

The objective of SunnyPro is to automate the preparation of sunny-side-up fried eggs as much as possible, making human interaction minimal for hotel breakfast areas, restaurants, and commercial kitchen environments. The machine stores raw eggs, separates and transfers them one by one, opens the eggs, removes the shells, transports the egg contents through a controlled heating zone, and delivers cooked eggs to the user. The system is designed to reduce manual labour, improve cooking consistency, and maintain hygienic separation between food material and waste material.

4.2. System Boundaries and Environmental Interaction

- System Boundary: The system boundary includes the internal mechanical, electrical, thermal, and control components of the machine. These include the egg storage unit, egg release mechanism, cracking mechanism, eggshell storage drawer, conveyor transport system, bottom heating system, top IR heating system, sensors, actuators, microcontroller, touchscreen interface, and safety components.

- External Environment: The external environment includes the user, the raw eggs added into the machine,the oil added to the machine , the cooked egg output , mains electrical power, ambient air, cleaning actions, and eggshell waste removal by the operator.

- Interactions:

The machine receives:

➢ Raw eggs from the operator
➢ Oil from the operator
➢ Electrical energy from the mains supply
➢ User input from the touchscreen
➢ Lining for Egg shell Storage Unit from Operator
➢ Operator settings input to the touchscreen
➢ Cleaning and maintenance actions from the operator

The machine outputs:

- ➢ Cooked sunny-side-up eggs
- ➢ Collected eggshell waste
- ➢ Heat losses to the surroundings
- ➢ Status information,Performance analytics, warnings, and error messages to the user

4.3. Working Principle (System Functionality Overview)

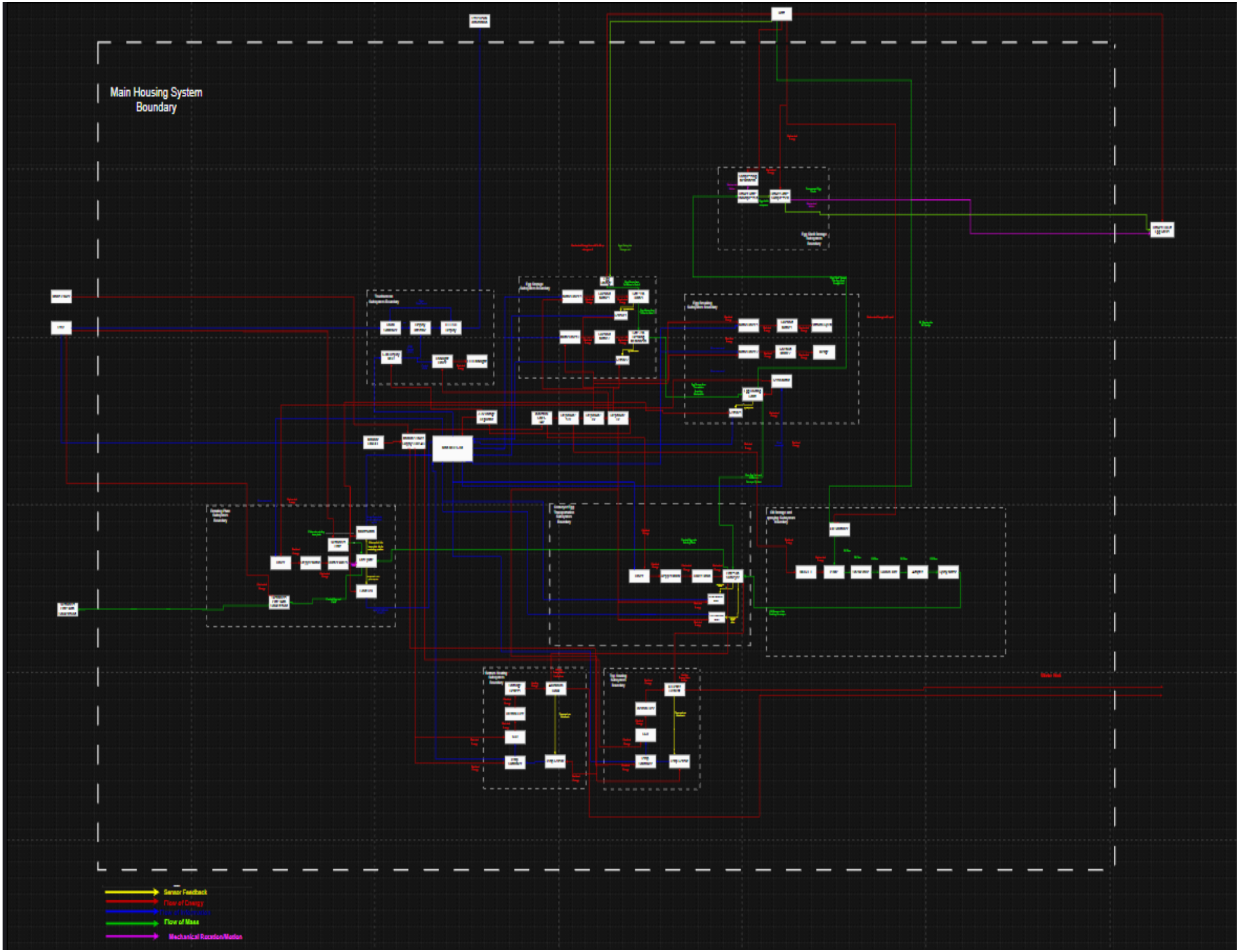
System Functionality Overview

1. **Preparation and Refilling:**
   Before operation, the kitchen staff fills the raw egg storage unit, checks and refills the oil container, inserts the eggshell storage drawer and drawer lining, and confirms that all removable food-contact parts are correctly installed.
2. **Startup:**
   The staff switches on the machine. The control system initializes the touchscreen, sensors, heating systems, conveyor, actuators, and safety components.
3. **Heating Settings / Doneness Selection:**
   The staff selects the required cooking setting through the touchscreen. This may include presets such as soft, medium, or well-cooked, or technical settings such as bottom heater temperature, top IR heater intensity, and conveyor speed. These settings determine how strongly and how long the egg is cooked.
4. **Preheating:**
   The bottom heating block and top IR heating system heat up to the selected operating temperature. Temperature sensors provide feedback to the controller. The machine only starts cooking when the required temperature range is reached.
5. **Customer / User Input:**
   The user selects the required number of eggs, from 1 to 4, through the touchscreen interface.The doneness setting is preset by staff for operation hours so as to not waste time waiting for temperature to go down or up between each user.
6. **Egg Storage and Release:**
   Raw eggs are stored in the egg storage unit. A gate release mechanism allows one egg at a time to move into the cracking position.
7. **Egg Detection:**
   Sensors confirm that an egg has reached the correct position before the cracking mechanism is activated.
8. **Egg Cracking:**
   The cracking mechanism opens the egg and separates the edible egg contents from the shell.
9. **Shell Removal:**
   The eggshells are directed into the eggshell storage drawer by the same breaking mechanism but through a different motor that basically tosses it to the side. The shell waste remains separated from the food path.

10. **Egg Transfer to Conveyor:**
    The egg contents are deposited into a removable egg pocket plate on the conveyor system.
11. **Oil Dosing / Surface Preparation:**
    A small amount of oil is supplied to the egg pocket automatically to reduce sticking and improve cooking quality. The oil container is refilled by the staff when required.
12. **Transport Through Heating Zone:**
    The conveyor moves the egg pocket through the cooking area at a controlled speed. The conveyor speed affects the cooking time.
13. **Bottom Heating:**
    The bottom heating block transfers heat to the underside of the egg pocket plate to cook the egg white from below.
14. **Top IR Heating:**
    The top IR heater applies radiant heat from above to support cooking of the upper egg surface while keeping the sunny-side-up appearance.
15. **Monitoring and Control:**
    Temperature sensors, home position sensors, egg detection sensors, oil level sensors, shell drawer position sensors, and safety cutoffs provide feedback to the control system.
16. **Egg Output:**
    The cooked egg leaves the heating zone and is transferred to the output area for the user or kitchen staff.
17. **Rotating Plate Positioning:**
    The rotating plate positions the cooked egg output correctly after each egg leaves the conveyor.
18. **Waste Removal and Refilling During Operation:**
    During longer operations, the kitchen staff can refill the raw egg storage unit and oil container. The eggshell storage drawer can be removed and emptied when the machine is paused or when the system indicates that the drawer is full.
19. **Shutdown / Standby:**
    The machine enters standby after the order is completed or shuts down if the staff ends service, or a safety fault is detected.

4.4. Functional Structure[17]

[17] Functional Structure Website.
https://www.drawio.com/


Main Housing System Boundary
Sensor Feedback
Flow of Energy
Flow of Mass
Mechanical Rotation/Motion

4.5. Functional Structure Description

4.5.1. User Interaction

The user interacts with SunnyPro through the touchscreen interface. The touchscreen allows the user to select the required number of fried eggs, from 1 to 4.The kitchen staff may also select the cooking level, such as soft, medium, or well-cooked.

The touchscreen sends the selected input as an information signal to the microcontroller. The microcontroller then starts the required sequence for egg release, cracking, cooking, transport, and output positioning. During operation, the touchscreen can also display machine status, warnings, heating readiness, egg storage status, oil level warnings, shell drawer status, and fault messages.

4.5.2. Operator Interaction and Preparation

Before operation, the kitchen staff prepares the machine for use. The staff fills the raw egg storage unit, checks or refills the oil container, inserts the eggshell storage drawer, and confirms that the removable egg pocket plates are correctly installed on the conveyor.

The staff can also adjust heating settings through the touchscreen. These settings affect the final cooking result of the egg. For example, the machine may change the bottom heater temperature, the top IR heater intensity, or the conveyor speed depending on the selected cooking level. A longer heating time or higher heating intensity produces a more cooked egg, while a shorter time or lower intensity produces a softer sunny-side-up result.

During longer operations, the staff can refill the raw egg storage unit and oil container. The eggshell drawer can also be removed and emptied when the system is paused or when the machine indicates that the drawer is full.

During machine initialization, the conveyor system also performs a positioning reset. When the machine is switched on, the microcontroller activates the conveyor motor and moves the conveyor slowly until the first egg pocket plate is detected by the home/reference sensor. This allows the control system to define the starting position of the conveyor before any egg is released. Once the first plate is correctly aligned under the cracking mechanism, the conveyor stops and the system confirms that the machine is ready for egg deposition. This initialization step ensures that the raw egg contents land directly into the pocket plate and prevents splashing, misalignment, or egg loss at the beginning of operation.

The conveyor positioning system uses sensors to ensure that the egg pocket plates are correctly aligned during operation. Since there are always four removable egg pocket plates mounted on the conveyor, the system must know the position of the plates before the egg is released from the cracking mechanism

4.5.3. Egg Storage and Release

Up-to 260 raw eggs can be stored in the egg storage unit before cooking. The storage unit holds the eggs in a controlled position so that the eggs can be released one at a time into the transfer tunnel. The release system uses two gates to separate the eggs and prevent more than one egg from reaching the cracking mechanism at the same time.

The first gate is positioned between the egg storage unit and the transfer tunnel. When the machine receives an order, the microcontroller sends a signal to the gate actuator driver. The driver activates the first gate for a short controlled time, allowing one egg to leave the storage unit and enter the transfer tunnel. After the egg passes through, the first gate closes again to block the next egg.

A sensor in the transfer tunnel detects whether an egg has entered correctly. This sensor sends a signal back to the microcontroller. If one egg is detected, the system continues the sequence. If no egg is detected, the microcontroller stops the process and displays an error message on the touchscreen. If the tunnel sensors detect that two eggs are present in the tunnel, the microcontroller identifies this as a double-feed condition and prevents the second gate from releasing the eggs into the cracking mechanism until the problem is corrected by the program logic.

The second gate is positioned between the transfer tunnel and the egg cracking mechanism. This gate remains closed while the first egg is being separated from the storage unit. Once the transfer tunnel sensor confirms that one egg is correctly positioned and the cracking mechanism is ready, the microcontroller sends a signal to the second gate actuator driver. The second gate opens and allows the egg to move from the tunnel into the cracking position.

A second egg detection sensor near the cracking platform confirms that the egg has reached the correct position on the egg cracking mechanism or arm. This signal is sent to the microcontroller. Only after this confirmation does the microcontroller activate the cracking sequence.

4.5.4. Egg Detection and Position Confirmation

The egg detection system confirms that the egg is correctly positioned before the cracking mechanism is activated. This is important because the cracking mechanism should not move if the egg is missing, misaligned, or stuck.

The sensor detects the egg on the cracking platform or egg holder. The sensor sends an information signal to the microcontroller. The microcontroller checks this signal in the program logic. If the egg is in the correct position, the microcontroller continues the cracking sequence. If the egg is not detected, the machine pauses and sends a warning to the user interface.

4.5.5. Egg Cracking Mechanism

The egg cracking mechanism opens the egg and separates the edible egg contents from the shell. When the sensor confirms that an egg is positioned on the cracking platform, it sends a signal to the microcontroller. The microcontroller then sends control signals to the solenoid drivers.

The Solenoid motors move the platform, spear, and cracking pin into the egg. The spear and the platform move with a controlled force enough to first penetrate the egg shell while the pin pushes through the spear sides to open up the egg at a controlled time The mechanism is strong enough to open the shell but controlled enough to avoid destroying the yolk.

After the shell is opened, the egg contents flow downward into the removable egg pocket plate on the conveyor. This is the main food mass flow.After the egg contents have been released, the cracking mechanism returns to its original position. After a short delay, the microcontroller sends a signal to the servo motor connected to the shell-removal shaft. The shaft slowly flicks the remaining eggshell waste into the eggshell tunnel.The shaft is directly mounted with metal horns.

4.5.6. Eggshell Collection System

The eggshell collection system receives the broken eggshells after the cracking process. After the egg contents have been deposited onto the conveyor, the shell-removal shaft flicks the shell waste into the eggshell tunnel. The tunnel guides the shells into the eggshell storage drawer simply through gravity because it is inclined.

The eggshell drawer is separated from the food path to reduce contamination risk. The drawer can be removed by the operator for emptying and cleaning. Inside the drawer, an inclined floor helps guide the eggshells toward the collection area.

The drawer also includes a manual pressing mechanism. The operator can press a handle connected to a pusher plate. This pusher plate compresses the eggshells inside the drawer, reducing the volume of waste and allowing the drawer to store more shells before emptying. Return springs bring the pressing mechanism back to its starting position after use.

A removable inner liner is placed inside the drawer. This liner prevents shell residue and liquid egg remains from sticking directly to the stainless-steel drawer surface. It also makes cleaning easier because the liner can be removed, washed, or replaced.

The egg shell drawer should be emptied out at the end of every day's use to prevent things like rot and unpleasant smells, The advised threshold for emptying the egg drawer is at max of 260 eggs ; however it has the capacity for over 500 egg shells when compressed.

4.5.7. Conveyor Transport System

The conveyor transport system moves the egg contents through the cooking zone. After the cracking mechanism releases the egg contents, the egg falls into a removable egg pocket plate mounted on the conveyor.

The pocket plates are designed to hold the egg in a controlled shape with the centre of it being the peak of the concavity and prevent the egg from spreading too far while keeping the yolk slightly closer and centred . Each plate is attached to the conveyor using mounting tabs and quick-release pins. These quick-release pins allow the plates to be removed for cleaning, maintenance, or replacement.

The conveyor is driven by a NEMA-17 stepper motor. The microcontroller sends step and direction signals to the stepper motor driver. The driver supplies the required current to the motor. The motor rotates the drive shaft through the coupling and sprocket system. The sprockets move the roller chains, and the chains carry the egg pocket plates through the machine.

A home or limit sensor gives the conveyor a reference position. This allows the microcontroller to know where the egg pockets are located. The position reference is important for aligning the pocket plates with the cracking position, heating zone, and output position.

The conveyor positioning system uses sensors to ensure that the egg pocket plates are correctly aligned during operation. Since there are always four removable egg pocket plates mounted on the conveyor, the system must know the position of the plates before the egg is released from the cracking mechanism.

A home/reference sensor is used to detect the first egg pocket plate before the raw egg contents are deposited. This ensures that the pocket plate is correctly positioned under the cracking mechanism so the egg lands inside the pocket instead of missing the plate or splashing outside the food path.

At the end of the conveyor, another position sensor detects when the last pocket plate has passed through the cooking and output area. When this final plate is detected, the sensor sends a signal to the microcontroller. The microcontroller then commands the conveyor motor to continue or reset the conveyor position until the first pocket plate reaches the home/reference position again.

4.5.8. Bottom Heating System

The bottom heating system cooks the egg from below. It consists of cartridge heaters installed in an aluminium heating block. The heating block is positioned below the conveyor path so that heat is transferred to the underside of the egg pocket plates.

The temperature sensor measures the temperature of the heating block. This temperature information is sent to the temperature controller and microcontroller with bi-directional communication. The controller compares the measured temperature with the selected set point. If more heat is needed, the controller sends a signal to the solid-state relay (SSR). The solid-state relay (SSR) switches power to the cartridge heaters.

The bottom heating system mainly affects the cooking of the egg white from below. It must provide stable heat without burning the egg or overheating the conveyor components. A thermal fuse or safety cutoff is included as protection. If the heating block exceeds a safe temperature, the safety device interrupts the heating circuit or sends a fault signal.

4.5.9. Top IR Heating System

The Top IR Heating System cooks the upper surface of the egg without flipping it. This is important because the machine is designed to produce sunny-side-up eggs, where the yolk should remain visible and intact.

The Top IR Heater emits radiant heat downward toward the egg surface. A reflector shield directs the infrared radiation toward the conveyor pockets and reduces heat loss. A guard cover protects the user and other components from direct contact with the hot heater.

The Top IR Heater is controlled by the microcontroller and temperature controller through a solid-state relay (SSR). The selected cooking setting determines how much top heat is applied. For a softer egg, the IR intensity or active time can be reduced. For a more cooked egg, the IR heater can remain active for longer or operate at a higher intensity.

A safety cutoff or temperature sensor monitors the upper heating zone. If the temperature becomes unsafe, the controller stops the heater and sends a warning to the Touchscreen.

4.5.10. Oil Dosing and Surface Preparation

The Oil System prepares the cooking surface so the egg does not stick to the pocket plate. Oil is stored in an oil container that can be refilled by the staff. Before egg deposition, a controlled amount of oil (2ml) is supplied to the egg pocket plate.

The oil is delivered by a small peristaltic pump. This improves cooking quality and reduces waste.

The Touchscreen then displays a refill warning for the staff as the amount of oil per egg is set so after the set amount of eggs are used the touchscreen displays and oil warning level.

4.5.11. Power and Control System

The machine is powered from the AC mains supply. The AC mains input is first connected to the main protection and switching components, such as the main switch, fuse, emergency stop circuit, grounding connection, and other required safety protection elements.

From the AC mains input, the power system is divided into two main paths. The first path supplies the heating systems, which remain AC-powered. The bottom heating cartridges and the top IR heater are not supplied through step-down buck converters. Instead, they are controlled through relays or solid-state relays (SSR), which switch the AC power to the heaters according to the temperature control signals.

The second path supplies the low-voltage control and actuator systems. The AC mains power is converted by the main DC power supply. From this DC supply, separate step-down buck converters and DC-DC converters provide the required voltage levels for the different subsystems. These voltage levels include 24 V, 12 V, 6 V, 5 V, and 3.3 V, depending on the component requirements.

Each subsystem is connected only to the voltage level it is designed for, which prevents components from being damaged by incorrect voltage.

The microcontroller acts as the central control unit. It receives information from the touchscreen and sensors, processes the program logic, and sends control signals to the motor drivers, solenoid drivers, servo drivers, relays, oil dosing system, conveyor system, and output system. The heaters use AC power, but their switching is still controlled by the low-voltage control system through the relay or solid-state relay (SSR) interface.

4.5.12. Output System

After the egg has passed through the heating zone, the conveyor reaches the output end. At this point, the cooked egg slides off the removable egg pocket plate and onto the rotating serving plate.

Before the output sequence starts, the system must first confirm that a serving plate is actually placed in the output area. A load cell is installed under the rotating plate platform. When the operator places a serving plate on the platform, the load cell detects the added weight and sends a signal to the microcontroller. This signal acts as the permission condition for the output system. If no plate is detected, the microcontroller does not allow the egg output sequence to begin and can display a warning on the touchscreen.

Once the load cell confirms that a plate is present, the microcontroller starts the rotating plate initialization. The rotating plate is driven by a stepper motor. The motor transfers motion through a transmission shaft to a rubber wheel, which rotates the plate by friction. The plate has a small bottom tab or reference feature. During initialization, the stepper motor rotates the plate slowly until this bottom tab hits or triggers a micro switch. When the micro switch is triggered, it sends a signal to the microcontroller. This defines the home position of the rotating plate.

The timing is controlled in the microcontroller code.After the home position is confirmed, the microcontroller can rotate the serving plate according to the number of eggs selected by the user. If the user selects two eggs, the plate rotates in 180° steps. If the user selects three eggs, it rotates in 120° steps. If the user selects four eggs, it rotates in 90° steps. This ensures that the eggs are distributed evenly on the serving plate.

For example:

2 eggs: $\frac{360°}{2}$ = 180° per step

3 eggs: $\frac{360°}{3}$ = 120° per step

4 eggs: $\frac{360°}{4}$ = 90° per step

This output system prevents multiple eggs from falling onto the same location and allows the final serving plate to be automatically arranged based on the selected order size.

4.6. Relation to D-FMEA

The functional structure supports the later D-FMEA because it shows the important functions, interfaces, and dependencies of the machine. For example, if the egg detection sensor fails, the cracking mechanism may operate at the wrong time. If the conveyor home sensor fails, the egg pocket may not align with the cracking or output position. If the bottom heating sensor fails, the egg may be undercooked or the heating block may overheat. If the shell chute becomes blocked, eggshell waste may contaminate the food path. By identifying these functional relationships early, the D-FMEA can focus on the most critical failure modes and define suitable prevention and detection measures.